

Investigation Into the Impact of Optimization-Based Control on Fault-Tolerant, $\mathcal{H}_2$ -Norm-Based Control

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1 Introduction

Fault-tolerant control (FTC) is a method of process control that is able to estimate and accommodate different types of faults within a system (actuator faults, sensor faults, plant faults, etc.). Allen and El-Farra have studied a form of FTC in their work exploring model-based, non-optimal, unconstrained control methods to use alongside fault estimation and accommodation [1]. Their work is heavily based on the work done by Montestruque and Antsaklis in their chapter within *Networked Embedded Sensing and Control*, which defines the Extended $\mathcal{H}_2$ -norm for continuous and discrete systems [2].

James' formulation of the control problem is to first estimate the magnitude of the actuator fault through moving-horizon estimation. Once the fault magnitude has been estimated, it is then used within an optimization problem that finds an adjusted fault parameter that minimizes the $\mathcal{H}_2$ -norm, which is a measure of how quickly the system stabilizes after an impulse disturbance. From the nature of $\mathcal{H}_2$ -norm calculation, it is quite similar to the calculation of the state and input cost for optimization-based control systems, which is where I'd like to explore.

My thinking is this: since the calculation of the $\mathcal{H}_2$ -norm is so similar to that of the cost function, there may be equivalencies such that if the system were designed with LQR in mind rather than a non-optimal model-based control method (like full-state feedback), then we can forgo the optimization that finds an adjusted fault magnitude, since LQR has already essentially done the optimization.

2 Terms and Definitions

2.1 Discrete Lyapunov Equation

The discrete Lyapunov equation is,

$$A^\top X A - X + Q = 0, \quad (2.1.1)$$

where X is a square symmetric matrix that solves the equation and $Q \succ 0$ that can be treated as a sort of tuning parameter. This equation describes the stability of a discrete autonomous system described by $x_{k+1} = Ax_k$. If $X \succ 0$ then the autonomous system is stable, otherwise it is unstable.

X can be found through an infinite summation:

$$X = \sum_{i=0}^{\infty} (A^\top)^i Q (A)^i, \quad (2.1.2)$$

which will become useful later on.

2.2 The Extended $\mathcal{H}_2$ -Norm

The extended $\mathcal{H}_2$ -norm is a performance measure of a discrete system that is an augmentation of the original system. In the context of fault-tolerant control, we must define a new state vector $\xi = [x^\top \ e^\top]^\top$ which is used in the autonomous system,

$$\dot{\xi} = \Lambda \xi, \quad (2.2.1)$$

where,

$$\Lambda = \begin{bmatrix} A + B\theta K & -B\theta K \\ (A - \hat{A}) + (B\theta - \hat{B}\hat{\theta})K & \hat{A} - (B\theta - \hat{B}\hat{\theta})K \end{bmatrix}. \quad (2.2.2)$$

As a reminder, we are considering the system,

$$\begin{aligned} \dot{x} &= Ax + B\theta u \\ \dot{\hat{x}} &= \hat{A}\hat{x} + \hat{B}\hat{\theta}u \\ u &= K\hat{x} \end{aligned} \quad (2.2.3)$$

where the hatted terms ($\hat{\cdot}$) are estimations of the actual parameter. This set of state-space models describes the behavior of the system without any disturbances, however if we wanted to know the impulse response then we'd need to instead study the system,

$$\begin{aligned} \dot{x} &= Ax + B\theta u + Ed \\ z &= Qx, \end{aligned} \quad (2.2.4)$$

where $d \in \mathbb{R}^u = \delta(t - t_0)$ is the impulse disturbance vector and $E \in \mathbb{R}^{u \times n}$ is the disturbance matrix. $z \in \mathbb{R}^h$ is the performance output and $Q \in \mathbb{R}^{n \times h} \succ 0$ is the performance matrix, which can be freely chosen to select specific states to be considered for performance determination.

With this new state-space model, we can form a new augmented system:

$$\begin{aligned} \dot{\xi} &= \Lambda \xi + Hd \\ \xi(t_j) &= [x^\top(t_j) \ 0]^\top, \ j \in \{0, 1, 2, 3, \dots\} \\ z &= N\xi, \end{aligned} \quad (2.2.5)$$

where $H = [E^\top \ 0]^\top$ and $N = [Q \ 0]$. We are interested in describing the system dynamics given a periodic sampling time $\Delta = t_{j+1} - t_j$, which can be found by solving equation (2.2.5) to yield,

$$z(t) = Ne^{\Lambda(t-t_j)} (I_0 e^{\Lambda\Delta})^j H, \ t \in [t_j, t_{j+1}), \quad (2.2.6)$$

After all this work, we are now able to describe the extended $\mathcal{H}_2$ -norm, which is defined as,

$$\|G\|_{\mathcal{H}_2} = \text{trace} (H^\top X H)^{1/2}, \quad (2.2.7)$$

where X is the solution to the discrete Lyapunov equation,

$$M(\Delta)^\top X M(\Delta) - X + W_o(0, \Delta) = 0, \quad (2.2.8)$$

with $M(\Delta) = I_0 e^{\Lambda \Delta}$ and $W_o(0, \Delta) = \int_0^\Delta e^{\Lambda^\top t} N^\top N e^{\Lambda t} dt$. Something potentially useful to note, from Montestruque and Antsaklis, is that:

$$\|G\|_{\mathcal{H}_2} = \|G\|_2 \quad (2.2.9)$$

given we set $A = M(\Delta)$, $B = H$, and $C = D = 0$.

3 Exploring the Math

With the definitions out of the way, I want to just explore around with the math and see what we find. I am really only going in the vague direction I laid out in the introduction, besides that I just want to see what I come up with.

3.1 $\mathcal{H}_2$ -Norm from 2-Norm

I must preface that I did not come up with this, this is copied from Montestruque and Antsaklis' work, but I found it interesting. We can obtain the expression for the extended $\mathcal{H}_2$ -norm by computing the 2-norm of the performance vector z ,

$$\begin{aligned} \|z\|_2^2 &= \int_{t_0}^{\infty} z^\top z dt \\ &= \int_{t_0}^{\infty} H^\top \left((I_0 e^{\Lambda \Delta})^\top \right)^j e^{\Lambda^\top (t-t_j)} N^\top N e^{\Lambda (t-t_j)} (I_0 e^{\Lambda \Delta})^j H dt \\ &= \sum_{i=0}^{\infty} \left(\int_0^\Delta H^\top \left((I_0 e^{\Lambda \Delta})^\top \right)^i e^{\Lambda^\top t} N^\top N e^{\Lambda t} (I_0 e^{\Lambda \Delta})^i H dt \right) \\ &= H^\top \sum_{i=0}^{\infty} \left(\left((I_0 e^{\Lambda \Delta})^\top \right)^i \left(\int_0^\Delta e^{\Lambda^\top t} N^\top N e^{\Lambda t} dt \right) (I_0 e^{\Lambda \Delta})^i \right) H, \end{aligned}$$

Let us define $M(\Delta) = I_0 e^{\Lambda \Delta}$ and $W_o(0, \Delta) = \int_0^\Delta e^{\Lambda^\top t} N^\top N e^{\Lambda t} dt$ to yield,

$$\|z\|_2^2 = H^\top \sum_{i=0}^{\infty} \left(\left((M(\Delta))^\top \right)^i W_o(0, \Delta) (M(\Delta))^i \right) H,$$

where we notice that the summation takes the form of the solution to the discrete Lyapunov equation,

$$X = \sum_{i=0}^{\infty} \left((M(\Delta))^\top \right)^i W_o(0, \Delta) (M(\Delta))^i$$

so the 2-norm of the performance output can be shortened to,

$$\|z\|_2^2 = H^\top X H \quad (3.1.1)$$

which is the basis for the $\mathcal{H}_2$ -norm (I'm not sure where the trace comes into play, but we're almost there).

3.2 Optimal $\hat{\theta}$ for LQR

Calculating the $\mathcal{H}_2$ -norm for a system controlled using LQR, and the controller gain is recalculated to become aware of θ , we find that the so-called “minimal accommodation strategy” where $\hat{\theta} = \bar{\theta}$ actually produces the lowest $\mathcal{H}_2$ -norm for LQR. See the figures below for the contour plots of the $\mathcal{H}_2$ -norm.

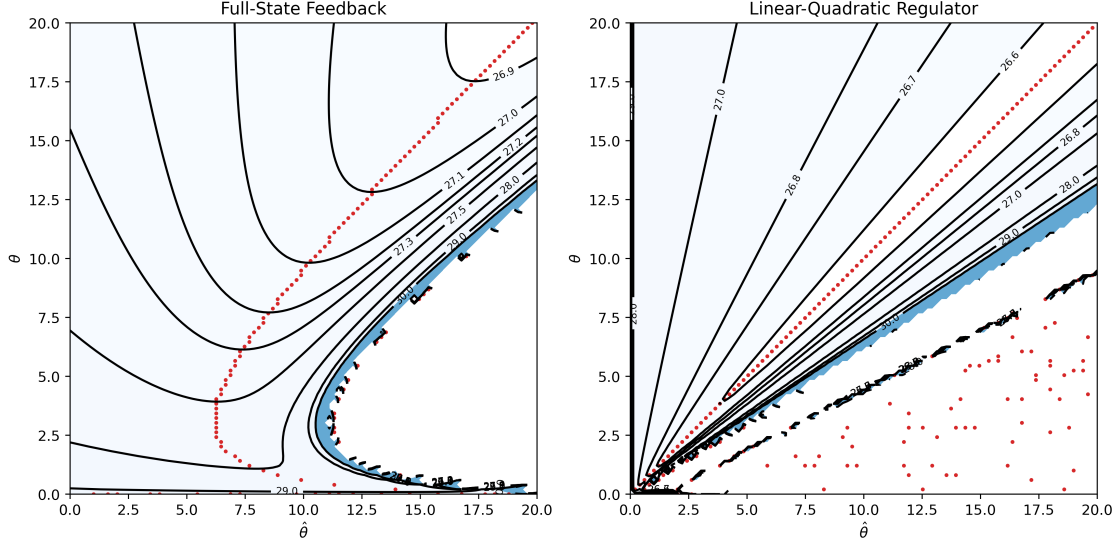


Figure 1: $\mathcal{H}_2$ -norm contour plots for full-state feedback (left) and LQR (right); the red dots indicate the choice of $\hat{\theta}$ that gives a local minimum for a given θ .

We can see from the 3D plot that there are choices of $\hat{\theta}$ that would provide a smaller $\mathcal{H}_2$ -norm than the minimal accommodation strategy, however they lie extremely close to the region of instability in parameter-space. It may be that the minimal accommodation strategy, if the system is using LQR, is the safest option. I was curious if the minimal accommodation strategy really does yield a local minimum in the $\mathcal{H}_2$ -norm, so I approximated the derivative along the $\theta = 10$ line just to see, which does show that there is a local minimum—see figure (2).

I also want to try to analytically find where $\hat{\theta}$ would be. We know that,

$$\hat{\theta}^* = \underset{\hat{\theta}}{\operatorname{argmin}} \{ \|G\|_{\mathcal{H}_2} \},$$

which could be found by seeing where $\nabla_{\hat{\theta}} \|G\|_{\mathcal{H}_2} = 0$, which does not require the Hessian to be positive definite (which would yield a global minimum), as a local minimum is also welcome. Let's first assume that the Hessian is indeed positive definite and find the gradient (I haven't even taken the gradient as of writing this part so I don't even know if this will work...):

$$\begin{aligned} \nabla_{\hat{\theta}} \|G\|_{\mathcal{H}_2} &= \nabla \|G\|_{\mathcal{H}_2} \\ &= \nabla \left(\operatorname{trace} (H^\top X H)^{1/2} \right) \\ &= \frac{1}{2} \operatorname{trace} (H H^\top X)^{-1/2} \operatorname{trace} (\nabla (H H^\top X)) \\ &= \frac{1}{2} \operatorname{trace} (H H^\top X)^{-1/2} \operatorname{trace} (H H^\top (\nabla X)^\top) \end{aligned}$$

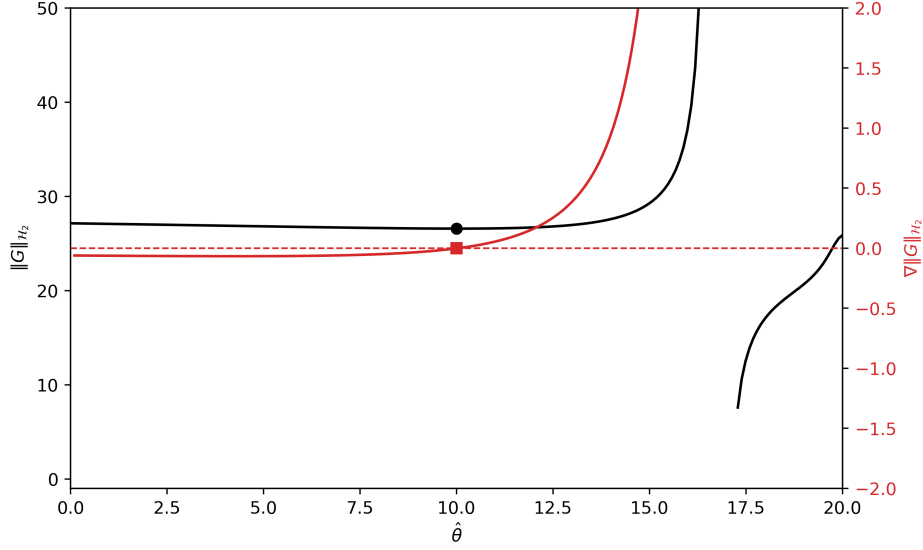


Figure 2: The numerical approximation of the derivative of the norm shows that $\hat{\theta} = \theta$ does indeed form a local minima. $\theta = 10$, the derivative is zero at $\hat{\theta} = 10$.

The only thing we need now is to find out what ∇X is. Recall that X is the solution to the discrete Lyapunov equation,

$$X = \sum_{i=0}^{\infty} \left((M(\Delta))^{\top} \right)^i W_o(0, \Delta) (M(\Delta))^i,$$

where we may be able to take its derivative:

$$\begin{aligned} \nabla X &= \nabla \left(\sum_{i=0}^{\infty} \left((M(\Delta))^{\top} \right)^i W_o(0, \Delta) (M(\Delta))^i \right) \\ &= \sum_{i=0}^{\infty} \left(\nabla \left((M(\Delta))^{\top} \right)^i W_o(0, \Delta) (M(\Delta))^i \right) \\ &= \nabla W_o + \sum_{i=1}^{\infty} \left[(\nabla(M^{\top}))^i W_o M^i + (M^{\top})^i (\nabla W_o) M^i + (M^{\top})^i W_o (\nabla M^i) \right] \end{aligned} \quad (3.2.1)$$

Let's find the derivatives for each term separately. First, $\nabla(M^{\top})^i$:

$$\begin{aligned} \nabla(M^{\top})^i &= i(M^{\top})^{i-1} \nabla(M^{\top}) \\ &= i(M^{\top})^{i-1} \cdot \frac{\partial}{\partial \hat{\theta}} \left(I_0 e^{\Lambda^{\top} \Delta} \right) \\ &= i(M^{\top})^{i-1} \cdot I_0 \int_0^1 e^{\alpha \Lambda^{\top} \Delta} \left(\Delta \frac{\partial \Lambda^{\top}}{\partial \hat{\theta}} \right) e^{(1-\alpha) \Lambda^{\top} \Delta} d\alpha, \end{aligned} \quad (3.2.2)$$

(derivative of matrix exponential taken from the Wikipedia page “Matrix exponential”) where,

$$\frac{\partial \Lambda}{\partial \hat{\theta}} = \begin{bmatrix} 0 & 0 \\ -\hat{B}K & \hat{B}K \end{bmatrix}, \quad (3.2.3)$$

and,

$$\frac{\partial \Lambda^\top}{\partial \hat{\theta}} = \begin{bmatrix} 0 & -\hat{B}K \\ 0 & \hat{B}K \end{bmatrix}, \quad (3.2.4)$$

(which unsurprisingly shows that $\partial_{\hat{\theta}}(\Lambda^\top) = (\partial_{\hat{\theta}}(\Lambda))^\top$). We can similarly say that ∇M^i is:

$$\nabla M^i = i(M)^{i-1} \cdot I_0 \int_0^1 e^{\alpha \Lambda \Delta} \left(\Delta \frac{\partial \Lambda}{\partial \hat{\theta}} \right) e^{(1-\alpha)\Lambda \Delta} d\alpha, \quad (3.2.5)$$

The last derivative is for $W_o(0, \Delta)$ and looks a little messy:

$$\begin{aligned} \nabla W_o(0, \Delta) &= \frac{\partial}{\partial \hat{\theta}} \int_0^\Delta e^{\Lambda^\top t} N^\top N e^{\Lambda t} dt \\ &= \int_0^\Delta \frac{\partial}{\partial \hat{\theta}} \left[e^{\Lambda^\top t} N^\top N e^{\Lambda t} \right] dt \\ &= \int_0^\Delta e^{\Lambda^\top t} N^\top N \frac{\partial}{\partial \hat{\theta}} [e^{\Lambda t}] + \frac{\partial}{\partial \hat{\theta}} [e^{\Lambda^\top t}] N^\top N e^{\Lambda t} dt \\ &= \int_0^\Delta \left[\int_0^1 e^{\alpha \Lambda^\top t} \left(t \frac{\partial \Lambda^\top}{\partial \hat{\theta}} \right) e^{(1-\alpha)\Lambda^\top t} d\alpha + \left(\int_0^1 e^{\alpha \Lambda t} \left(t \frac{\partial \Lambda}{\partial \hat{\theta}} \right) e^{(1-\alpha)\Lambda t} d\alpha \right)^\top \right] N^\top N e^{\Lambda t} dt \end{aligned} \quad (3.2.6)$$

Okay, so we may have an analytical form for the derivative, good. The problem is that it doesn't really do us any good, at least not from my viewpoint. The form of the derivative doesn't seem to give us any benefit over using the numerical derivative, save the fact that we can analytically describe it and not rely on numerical methods.

4 Conclusion

This section is shorter than I would've hoped (I wanted to have a more fleshed-out exploration of the optimization and to see where we could reduce the number optimization steps done by using LQR) but for now I think this is all I could do with the time I have. I initially wanted to put these findings into a paper to submit to the American Control Conference (ACC) but it's much too unfinished for that. I'll keep going down the rabbit hole into how LQR is nearly identical to $\mathcal{H}_2$ -norm control, but for now I'll keep it at this.

The cool thing that we were able to find, however, is that the optimal choice of $\hat{\theta}$ using LQR is the $\hat{\theta} = \theta$ line, and for full-state feedback it appears to almost converge onto the $\hat{\theta} = \theta$ line, however we have no actual proof that this is the case. This helps to confirm my thinking that optimality plays a huge role in the choice of $\hat{\theta}$, and this finding matches those that we found analyzing the $\mathcal{H}_2$ -norm using MPC.

References

- [1] J. T. Allen and N. H. El-Farra, "A model-based framework for fault estimation and accommodation applied to distributed energy resources," *Renewable Energy*, vol. 100, p. 35–43, Jan 2017.
- [2] L. A. Montestruque and P. J. Antsaklis, *Performance Evaluation for Model-Based Networked Control Systems*, pp. 231–249. Berlin, Heidelberg: Springer Berlin Heidelberg, 2006.